Example 3.19

Obtain the condition that the roots of $x^3 + px^2 + qx + r = 0$ are in A.P.

Solution

Let the roots be in A.P. Then, we can assume them in the form $\alpha - d, \alpha, \alpha + d$.

Applying the Vieta's formula $(\alpha - d) + \alpha + (\alpha + d) = -\frac{p}{1} = p \Rightarrow 3\alpha = -p \Rightarrow \alpha = -\frac{p}{3}$.

But, we note that α is a root of the given equation. Therefore, we get

$$\left(-\frac{p}{3}\right)^3 + p\left(-\frac{p}{3}\right)^2 + q\left(-\frac{p}{3}\right) + r = 0 \Rightarrow 9pq = 2p^3 + 27r .$$

Example 3.20

Find the condition that the roots of $ax^3 + bx^2 + cx + d = 0$ are in geometric progression. Assume $a, b, c, d \neq 0$

Solution

Let the roots be in G.P.

Then, we can assume them in the form $\frac{\alpha}{\lambda}, \alpha, \alpha\lambda$.

Applying the Vieta's formula, we get

$$\sum_1 = \alpha\left(\frac{1}{\lambda} + 1 + \lambda\right) = -\frac{b}{a} \quad \ldots (1)$$

$$\sum_2 = \alpha^2\left(\frac{1}{\lambda} + 1 + \lambda\right) = \frac{c}{a} \quad \ldots (2)$$

$$\sum_3 = \alpha^3 = -\frac{d}{a}. \quad \ldots (3)$$

Dividing (2) by (1), we get

$$\alpha = -\frac{c}{b} \quad \ldots (4)$$

Substituting (4) in (3), we get $\left(-\frac{c}{b}\right)^3 = -\frac{d}{a} \Rightarrow ac^3 = db^3$.

Example 3.21

If the roots of $x^3 + px^2 + qx + r = 0$ are in H.P. , prove that $9pqr = 27r^2 + 2q^3$.

Assume $p, q, r \neq 0$

Solution

Let the roots be in H.P. Then, their reciprocals are in A.P. and roots of the equation

$$\left(\frac{1}{x}\right)^3 + p\left(\frac{1}{x}\right)^2 + q\left(\frac{1}{x}\right) + r = 0 \Leftrightarrow rx^3 + qx^2 + px + 1 = 0 . \quad \ldots (1)$$

Since the roots of (1) are in A.P., we can assume them as $\alpha - d, \alpha, \alpha + d$.

Applying the Vieta's formula, we get

$$\sum_1 = (\alpha - d) + \alpha + (\alpha + d) = -\frac{q}{r} \Rightarrow 3\alpha = -\frac{q}{r} \Rightarrow \alpha = -\frac{q}{3r}.$$

Chapter 3 Theory of Equation.indd 116 31-01-2020 17:32:37